

NYCU IMF - 114 Financial Econometrics

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Assignment: 9, Ch15 q6(f), 20(def)

Question 6 (f)

Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10		Estimation Results for Exercise 15.6			
	(1)	(2)	(3)	(4)	(5)
	OLS 1987	OLS 1988	FE	FE Robust	RE
C	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
$EXPER$	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
$EXPER^2$	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
$SOUTH$	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
$UNION$	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
N	716	716	1432	1432	1432

(standard errors in parentheses)

Figure 1: Table 15.10

- (f) Column (5) contains the random effect estimates. Which coefficients, apart from the intercepts, show them the most difference from the fixed effect estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

Random Effects vs. Fixed Effects (The Hausman Test)

1. Which coefficients show the most difference? When evaluating the percentage change from the consistent Fixed Effects estimates (Column 3) to the Random Effects estimates (Column 5), the coefficients related to experience show the most extreme

differences. The linear return to experience (*EXPER*) increases by 71.48%, and the magnitude of the quadratic penalty (*EXPER*²) almost doubles, changing by 91.67%. While *SOUTH* and *UNION* also shift, their changes are much less severe (shrinking by 28.67% and growing by 24.94%, respectively). Therefore, *EXPER* and *EXPER*² are the most sensitive to the model specification.

2. "Why that one?" (Why compare to Column 3 instead of Column 4?)

The question asks why we specifically compare the RE estimates to the standard FE estimates in Column (3), rather than the robust ones in Column (4).

The classic Hausman test relies on a strict theoretical assumption: under the null hypothesis (that unobserved effects are uncorrelated with regressors), the Random Effects estimator is fully efficient. This implies that the idiosyncratic errors are perfectly homoskedastic and have no serial correlation. The standard errors in Column (3) are derived under these exact same strict assumptions.

If we were to use the robust standard errors from Column (4), we would be admitting that the errors are heteroskedastic or serially correlated. If that were true, RE would no longer be fully efficient, and the mathematical simplification that makes the classic Hausman test formula (15.36) work would completely fall apart.

3. Calculating the Hausman *t*-statistic (Equation 15.36)

The formula for the Hausman *t*-test for an individual coefficient is:

$$t = \frac{\hat{\beta}_{FE} - \hat{\beta}_{RE}}{\sqrt{SE(\hat{\beta}_{FE})^2 - SE(\hat{\beta}_{RE})^2}}$$

Let's test the variable that showed a highly noticeable shift: *EXPER*.

- $\hat{\beta}_{FE} = 0.0575$
- $\hat{\beta}_{RE} = 0.0986$ $SE(\hat{\beta}_{FE}) = 0.0330$ $SE(\hat{\beta}_{RE}) = 0.0220$

$$t = \frac{0.0575 - 0.0986}{\sqrt{0.0330^2 - 0.0220^2}}$$

$$t = \frac{-0.0411}{\sqrt{0.001089 - 0.000484}}$$

$$t = \frac{-0.0411}{\sqrt{0.000605}}$$

$$t = \frac{-0.0411}{0.024597} \approx -1.67$$

4. Conclusion: Is Random Effects appropriate?

Statistical Conclusion: The calculated *t*-statistic for *EXPER* is $|-1.67|$. Because this does not exceed the standard 5% critical value of 1.96, we technically fail to reject the

null hypothesis at the 5% significance level. From a strictly statistical standpoint using this specific test, there is not enough evidence to definitively say the RE estimates are biased, meaning the RE model could be considered appropriate.

Question 20 (def)

This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.

- (d) Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.

School Random Effects Estimation

1. Comparison of Results (OLS vs. FE vs. RE)

Table 1: Comparison of OLS, Fixed Effects, and Random Effects Estimates

Variable	OLS (Part a)	FE (Part b)	RE (Part d)
<i>SMALL</i>	5.823***	6.490***	6.459***
<i>AIDE</i>	0.818	0.996	0.992
<i>TCHEXPER</i>	0.492***	0.286***	0.303***
<i>BOY</i>	−6.156***	−5.456***	−5.512***
<i>WHITE_ASIAN</i>	3.906***	8.028***	7.350***
<i>FREELUNCH</i>	−14.771***	−14.594***	−14.584***

Note: *** indicates $p < 0.001$.

Econometrically, the Random Effects (RE) estimator is a matrix-weighted average of the pooled OLS estimator and the Fixed Effects (FE) estimator. Looking at the results, this theoretical property holds true: the RE coefficients mathematically fall between the OLS and FE estimates. For example, the coefficient for *TCHEXPER* (0.303) sits squarely between the FE estimate (0.286) and the OLS estimate (0.492).

2. Variables Correlated with School Effects

The fundamental assumption necessary for the RE model to be unbiased is that the unobserved school effects (u_i) are strictly uncorrelated with all explanatory variables (i.e., $Cov(x_{it}, u_i) = 0$).

In this education context, several variables almost certainly violate this assumption:

- *FREELUNCH* and *WHITE_ASIAN*: These student demographic factors are deeply tied to residential sorting. Neighborhood wealth directly impacts both the demographic makeup of a school and the unobserved school characteristics (u_i), such as local property tax funding, quality of facilities, and positive peer effects.
- *TCHEXPER*: Highly desirable, well-resourced schools (a positive u_i) generally have a much easier time attracting and retaining highly experienced teachers compared to underfunded schools.

Because these variables are naturally correlated with school-level unobservables, the strict zero-correlation assumption of the RE model is highly suspect here.

3. The LM Test for Random Effects

To formally test whether random school effects exist in the data (compared to having no school effects at all), we use the Breusch-Pagan LM test.

- Null Hypothesis (H_0): The variance of the unobserved school-level effects is zero ($\sigma_u^2 = 0$). This implies there is no significant unobserved heterogeneity across schools, and standard pooled OLS is perfectly adequate.
 - Test Results: The test yields a chi-squared statistic of 6677.4 with a p-value of $< 2.2 \times 10^{-16}$.
 - Conclusion: Because the p-value is effectively zero, we overwhelmingly reject the null hypothesis. There is highly significant unobserved heterogeneity across schools. This confirms that the Random Effects model is strongly preferred over the standard pooled OLS model (though the upcoming Hausman test will determine if it is preferred over Fixed Effects).
- (e) Using the t-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

1. The Theoretical Setup (Equation 15.36)

The standard Hausman test is a joint test of all coefficients. However, equation (15.36) refers to a t -test used to evaluate the difference for a single coefficient.

Under the null hypothesis that Random Effects is consistent (i.e., unobserved school effects are uncorrelated with the regressors), the RE estimator is perfectly efficient, and the variance of the difference between the FE and RE estimates simplifies gracefully.

The test statistic is calculated as:

$$t = \frac{\hat{\beta}_{FE} - \hat{\beta}_{RE}}{\sqrt{SE(\hat{\beta}_{FE})^2 - SE(\hat{\beta}_{RE})^2}}$$

2. Calculating the t -statistics

Using the estimates and standard errors from the previous outputs (Part b and Part d), we calculate the test statistic for each requested variable. (Note: Because we have a very large sample size, the critical value for a 5% significance level is $\approx \pm 1.96$.)

- *SMALL*:
 - Diff: $6.490 - 6.459 = 0.031$
 - SE of Diff: $\sqrt{0.9129^2 - 0.9125^2} \approx 0.0275$
 - t -stat: $0.031/0.0275 = \mathbf{1.14}$ (Not significant)
- *AIDE*:
 - Diff: $0.996 - 0.992 = 0.004$
 - SE of Diff: $\sqrt{0.8817^2 - 0.8812^2} \approx 0.0307$
 - t -stat: $0.004/0.0307 = \mathbf{0.13}$ (Not significant)
- *TCHEXPER*:
 - Diff: $8.028 - 7.350 = 0.678$
 - SE of Diff: $\sqrt{1.5356^2 - 1.4313^2} \approx 0.5562$
 - t -stat: $0.678/0.5562 = \mathbf{1.22}$ (Not significant)
- *WHITE_ASIAN*:
 - Diff: $6.490 - 6.459 = 0.031$
 - SE of Diff: $\sqrt{0.9129^2 - 0.9125^2} \approx 0.0275$
 - t -stat: $0.031/0.0275 = \mathbf{1.14}$ (Not significant)
- *FREELUNCH*:
 - Diff: $-14.594 - (-14.584) = -0.010$
 - SE of Diff: $\sqrt{0.8800^2 - 0.8747^2} \approx 0.0967$
 - t -stat: $-0.010/0.0967 = \mathbf{-0.10}$ (Not significant)

3. Implications of the Test Outcomes

None of the individual t -statistics exceed the critical value of 1.96. This implies that, when looking at these coefficients one by one, there is no statistically significant difference between the Fixed Effects and Random Effects estimates at the 5

4. What happens with the coefficient on *BOY*?

If we try to apply this exact formula to *BOY*, the math breaks down.

- $SE(\hat{\beta}_{FE}) = 0.727589$
- $SE(\hat{\beta}_{RE}) = 0.727639$
- Variance Difference: $0.727589^2 - 0.727639^2 = \mathbf{-0.00007}$

We cannot take the square root of a negative number. Asymptotically, the variance of the FE estimator should always be greater than or equal to the variance of the efficient RE estimator. However, in finite samples with unbalanced panels (which the

star dataset is), it is mathematically possible for the FE standard error to be slightly smaller than the RE standard error. When this happens, the t -test statistic cannot be computed.

- (f) Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

The Mundlak Test

1. The Method and Hypotheses

To carry out the Mundlak test, we first created school-level averages for all explanatory variables (e.g., `mean_small`, `mean_freelunch`). We then estimated an auxiliary model that includes both the original variables and these newly created school-averages.

The test evaluates whether these school-averages are jointly significant.

- Null Hypothesis (H_0): The coefficients on all the school-average variables are jointly equal to zero. This implies that the unobserved school-level effects (u_i) are uncorrelated with the explanatory variables, meaning the Random Effects (RE) estimator is consistent and efficient.
- Alternative Hypothesis (H_1): The coefficients are not jointly equal to zero. This implies that the unobserved school effects are correlated with the regressors, meaning the RE estimator is biased and the Fixed Effects (FE) estimator must be used.

2. Test Results

Using a joint linear hypothesis test (an F -test) on the auxiliary regression, we obtained the following results:

- F-statistic: 21.004
- p-value: $< 2.2e - 16$

3. Final Conclusion

Because the p-value is effectively zero (vastly smaller than 0.05), we overwhelmingly reject the null hypothesis.

There is highly significant statistical evidence that the unobserved school-level heterogeneity (such as school funding, principal quality, or neighborhood wealth) is correlated with the observed explanatory variables (such as class size, teacher experience, and student demographics). Because this fundamental assumption of the Random Effects model is violated, the Random Effects estimates are inconsistent. Therefore, we conclude that the Fixed Effects model estimated in Part (b) is the appropriate and reliable model for analyzing the STAR experiment data.

Table 2: Coefficient Estimates from the Mundlak Auxiliary Regression

Variable	Individual Level (X_{it})	School-Level Mean ($\bar{X}_i$)
<i>SMALL</i>	6.483*** (0.995)	-23.103*** (5.795)
<i>AIDE</i>	1.041 (0.961)	9.187 [†] (5.238)
<i>TCHEXPER</i>	0.285*** (0.077)	0.802*** (0.179)
<i>BOY</i>	-5.461*** (0.793)	-50.212*** (7.119)
<i>WHITE_ASIAN</i>	8.043*** (1.674)	-7.221** (2.331)
<i>FREELUNCH</i>	-14.591*** (0.959)	-4.116 (2.510)

Note: Standard errors are reported in parentheses below the coefficient estimates. Significance levels: [†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The constant term (465.069***) is omitted for brevity.